

L5b: Minimum-Variance Portfolios, the Efficient Frontier, and the Capital Allocation Line

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Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

By the end of this lecture, you should be able to:

- **Formulate portfolio reward and risk:** derive expected growth and variance from weighted asset growth rates in matrix form; explain diversification through covariance.
- **Construct minimum-variance portfolios:** derive global minimum-variance weights by Lagrange multipliers, allowing shorts; formulate equality/inequality growth targets, distinguish full and efficient frontiers, and explain long-only constraints.
- **Add a risk-free asset:** derive the capital allocation line and tangent weights with assumptions; explain Sharpe ratios and two-fund separation under borrowing restrictions or unequal lending/borrowing rates.

Lectures and Examples

Lecture: Minimum-Variance Portfolios and the Capital Allocation Line

Example 1: Minimum-Variance Portfolios from Data

How do we choose portfolio weights from historical data? We estimate inputs from 2014–2024 data, solve the minimum-variance problem, trace the frontier and capital allocation line, and compare portfolios using 2025 prices.

Example 2: Portfolio Wealth with Multiple Asset GBM

How much might a chosen portfolio's wealth vary? We simulate correlated prices and buy-and-hold wealth, compare with 2025 observations, and estimate the probability of exceeding a target scaled net present value.

Concept Review: Multiple Asset GBM

Recall the multiple asset geometric Brownian motion model from L5a. With constant parameters, the price of asset $i = 1, \dots, M$ advances by:

$$S_i(t + \Delta t) = S_i(t) \exp \left[\mu_{g,i} \Delta t + \sqrt{\Delta t} (\mathbf{A}\mathbf{Z})_i \right].$$

Here, $\Delta t > 0$ is measured in years, and:

- $\mu_{g,i}$ is the asset's **mean growth rate** (year⁻¹).
- The loading matrix satisfies $\mathbf{A}\mathbf{A}^\top = \mathbf{C}$, where $\mathbf{C}$ is the **covariance rate** (year⁻¹).
- Use the same vector $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_M)$ for all assets in a step; draw a new independent vector at each step.

We estimate the mean growth rates and covariance rate from historical growth observations.

Concept Review: Estimating Covariance

From $N \geq 2$ aligned growth-rate observations, $\mathbf{g}'$ estimates the mean vector μ_g .

Subtract each asset's sample mean to form $\tilde{\mathbf{G}} \in \mathbb{R}^{N \times M}$.

Recall $\Sigma_g = \mathbf{C}/\Delta t$. The corresponding estimates are:

$$\boxed{\hat{\Sigma}_g = \frac{\tilde{\mathbf{G}}^\top \tilde{\mathbf{G}}}{N - 1}}, \quad \boxed{\hat{\mathbf{C}} = \Delta t \hat{\Sigma}_g}.$$

These estimates have different roles:

- Use $\hat{\Sigma}_g$ to calculate portfolio growth-rate variance (year^{-2}) and $\hat{\mathbf{C}}$ to simulate prices (covariance rate: year^{-1}).
- The sample covariance is positive semidefinite, so variances are nonnegative. Today's closed-form weights require a positive-definite estimate.

L5a example: [Covariance and Correlation from Data](#). Continue here if unfinished.

Concept Review: Wealth and Scaled NPV

Initial weights $w_i \geq 0$, with $\sum_i w_i = 1$, divide wealth $W_0 > 0$ among the assets. Buy $n_i = w_i W_0 / S_i(0)$ shares and hold them fixed:

$$W_t = W_0 \sum_{i=1}^M w_i \frac{S_i(t)}{S_i(0)},$$

$$\rho_T = \frac{W_T}{W_0} e^{-g_y T} - 1.$$

Here, g_y is the constant benchmark growth rate and T is the holding period in years. We allow fractional shares and omit dividends and trading costs.

Simulated wealth outcomes estimate $\mathbb{P}(\rho_T > \rho_\star)$ for a target scaled NPV $\rho_\star$.

L5a sampled candidate allocations. Today, we minimize variance for a required expected growth rate.

Modern Portfolio Theory: Portfolio Reward

For M risky assets, w_i is the initial fraction invested in asset i , with $\sum_i w_i = 1$. Hold the weights fixed when calculating reward and risk. The model uses weighted growth rates over one observation interval:

$$g_p = \sum_{i=1}^M w_i g_i = \mathbf{w}^\top \mathbf{g}.$$

With fixed weights and $\mathbb{E}[g_i] = \mu_{g,i}$, **reward** is:

$$\mathbb{E}[g_p] = \sum_{i=1}^M w_i \mathbb{E}[g_i] = \mathbf{w}^\top \boldsymbol{\mu}_g.$$

From data, $\mathbf{w}^\top \mathbf{g}'$ estimates expected growth (year^{-1}).

Expected growth is linear in the portfolio weights.

Weighted Growth and Buy-and-Hold Wealth

With fixed share counts, one-period wealth and its log growth rate are given by:

$$\frac{W_{\Delta t}}{W_0} = \sum_{i=1}^M w_i e^{g_i \Delta t},$$

$$\frac{1}{\Delta t} \ln \left(\frac{W_{\Delta t}}{W_0} \right) = \frac{1}{\Delta t} \ln \left(\sum_{i=1}^M w_i e^{g_i \Delta t} \right), \quad W_{\Delta t} > 0.$$

The logarithm acts on a sum, so this generally differs from $g_p = \sum_i w_i g_i$.

Use weighted growth rates to choose weights, then use buy-and-hold wealth to evaluate the investment through time.

Next, we measure how the weighted growth rate varies across outcomes.

Modern Portfolio Theory: Portfolio Risk

We measure **risk** by $\text{Var}(g_p)$. With fixed weights and covariance matrix $\Sigma_g = \text{Cov}(\mathbf{g})$:

$$\begin{aligned}\text{Var}(g_p) &= \text{Cov}\left(\sum_i w_i g_i, \sum_j w_j g_j\right) \\ &= \sum_i \sum_j w_i w_j \text{Cov}(g_i, g_j) = \mathbf{w}^\top \Sigma_g \mathbf{w}.\end{aligned}$$

Individual variances (diagonal entries) and pairwise covariances (off-diagonal entries) both contribute to risk. From data, use $\hat{\Sigma}_g$.

Variance has units year^{-2} . Its square root, $\sigma_{g,p} = \sqrt{\text{Var}(g_p)}$, is the growth-rate standard deviation (year^{-1}).

To see how covariance can reduce risk, consider a two-asset portfolio.

Two-Asset Diversification

Invest fractions w and $1 - w$, with $0 < w < 1$, in two assets with positive standard deviations $\sigma_{g,1}$ and $\sigma_{g,2}$ and correlation ρ_{12} :

$$\text{Var}(g_p) = w^2\sigma_{g,1}^2 + (1 - w)^2\sigma_{g,2}^2 + 2w(1 - w)\rho_{12}\sigma_{g,1}\sigma_{g,2}.$$

Correlation determines what mixing achieves:

- At $\rho_{12} = 1$, the standard deviation is the weighted average:
 $\sigma_{g,p} = w\sigma_{g,1} + (1 - w)\sigma_{g,2}.$
- For $\rho_{12} < 1$, it falls below that weighted average.
- If ρ_{12} is below the smaller standard deviation divided by the larger, some allocation has less variance than either asset. For 2 and 4 per year, the cutoff is $\rho_{12} < 0.5$.

Positive correlation can still permit diversification.

Growth Rates and Log Returns

For an observation interval $\Delta t > 0$, asset log returns are $\mathbf{r} = \Delta t \mathbf{g}$. Their weighted average is $r_p = \mathbf{w}^\top \mathbf{r} = \Delta t g_p$:

$$\mathbb{E}[r_p] = \Delta t \mathbb{E}[g_p],$$

$$\text{Var}(r_p) = \Delta t^2 \text{Var}(g_p).$$

Log returns are dimensionless, with covariance $\Sigma_r = \Delta t^2 \Sigma_g$.

Scaling preserves the optimization problem:

- Every portfolio variance is multiplied by the same positive factor Δt^2 , preserving the minimizing weights.
- Scale the target as $r_\star = \Delta t g_\star$ so that the same portfolios meet the target.

Growth rates and log returns therefore give the same optimal weights.

Standard Deviation and Volatility

Growth-rate standard deviation depends on the observation interval. To calculate volatility, use the covariance rate $\mathbf{C} = \Delta t \Sigma_g$ from L5a:

$$\sigma_{g,p} = \sqrt{\mathbf{w}^\top \Sigma_g \mathbf{w}} \quad (\text{year}^{-1}),$$

$$\text{Volatility} = \sqrt{\mathbf{w}^\top \mathbf{C} \mathbf{w}} = \sqrt{\Delta t} \sigma_{g,p} \quad (\text{year}^{-1/2}).$$

For daily observations, $\Delta t = 1/252$ years:

$$\hat{\mathbf{C}} = \frac{\hat{\Sigma}_g}{252}.$$

The **risk axis** in this lecture shows $\sigma_{g,p}$, measured in year^{-1} .

With reward and risk defined, we can now choose the weights that minimize variance.

The Global Minimum-Variance Portfolio

We seek the fully invested portfolio with the smallest growth-rate variance. **Short positions are allowed** ($w_i < 0$), and there is **no expected-growth constraint**. Assume Σ_g is symmetric and positive definite.

With $\mathbf{1} \in \mathbb{R}^M$ the vector of ones, the problem is:

$$\begin{aligned} & \underset{\mathbf{w} \in \mathbb{R}^M}{\text{minimize}} && \frac{1}{2} \mathbf{w}^\top \Sigma_g \mathbf{w} \\ & \text{subject to} && \mathbf{1}^\top \mathbf{w} = 1. \end{aligned}$$

The solution and its variance are:

$$\mathbf{w}_{\text{GMV}} = \frac{\Sigma_g^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma_g^{-1} \mathbf{1}}, \quad \sigma_{g,\text{GMV}}^2 = \frac{1}{\mathbf{1}^\top \Sigma_g^{-1} \mathbf{1}}.$$

The weights depend only on covariance. We now derive this result.

Deriving the GMV Weights

Introduce a multiplier λ for the budget constraint. The factor $1/2$ simplifies the derivative without changing the minimizing weights:

$$\mathcal{L}(\mathbf{w}, \lambda) = \frac{1}{2} \mathbf{w}^\top \Sigma_g \mathbf{w} - \lambda (\mathbf{1}^\top \mathbf{w} - 1).$$

- **Stationarity:** $\Sigma_g \mathbf{w} - \lambda \mathbf{1} = \mathbf{0}$, so $\mathbf{w} = \lambda \Sigma_g^{-1} \mathbf{1}$.
- **Budget:** $1 = \lambda \mathbf{1}^\top \Sigma_g^{-1} \mathbf{1}$, giving $\lambda = (\mathbf{1}^\top \Sigma_g^{-1} \mathbf{1})^{-1}$.
- **Variance:** use $\Sigma_g \mathbf{w} = \lambda \mathbf{1}$ and $\mathbf{w}^\top \mathbf{1} = 1$:

$$\sigma_{g,\text{GMV}}^2 = \mathbf{w}^\top \Sigma_g \mathbf{w} = \lambda \mathbf{w}^\top \mathbf{1} = \lambda = \frac{1}{\mathbf{1}^\top \Sigma_g^{-1} \mathbf{1}}.$$

Positive definiteness makes the objective strictly convex, so these weights give the unique minimum.

The Long-Only Problem

Long-only weights are nonnegative and sum to one. We also require estimated expected growth to reach a **target floor** $g_\star$ (year^{-1}).

Using sample means $\mathbf{g}'$ and estimated covariance $\hat{\Sigma}_g$, solve:

$$\begin{aligned} & \underset{\mathbf{w} \in \mathbb{R}^M}{\text{minimize}} && \mathbf{w}^\top \hat{\Sigma}_g \mathbf{w} \\ & \text{subject to} && (\mathbf{g}')^\top \mathbf{w} \geq g_\star, \\ & && \mathbf{1}^\top \mathbf{w} = 1, \\ & && 0 \leq w_i \leq 1, \quad i = 1, \dots, M. \end{aligned}$$

This is a **convex quadratic program**: the covariance matrix is positive semidefinite and the constraints are linear.

How do these constraints change the solution?

Adding the Portfolio Constraints

Use the same estimated inputs and add the restrictions in order:

1. **First, exclude short positions.** If the original GMV weights are all nonnegative, keep them. Otherwise, solve numerically with $w_i \geq 0$ and $\mathbf{1}^\top \mathbf{w} = 1$. The result is the **long-only GMV portfolio**.
2. **Then, impose the growth floor.** If the long-only GMV portfolio meets g_* , keep its weights. For a higher feasible target, solve again; the weights change and minimum variance increases.

At either step, restricting the portfolio choices cannot lower the minimum variance.

The Target-Growth Frontier

Allow short positions and require $\mathbf{1}^\top \mathbf{w} = 1$. For each **equality target** $\mu_g^\top \mathbf{w} = g_\star$, find the weights with the smallest variance.

Assume Σ_g is positive definite and the asset means are not all equal. Define:

$$\begin{aligned} a &= \mathbf{1}^\top \Sigma_g^{-1} \mathbf{1}, & b &= \mathbf{1}^\top \Sigma_g^{-1} \mu_g, \\ c &= \mu_g^\top \Sigma_g^{-1} \mu_g, & d &= ac - b^2 > 0. \end{aligned}$$

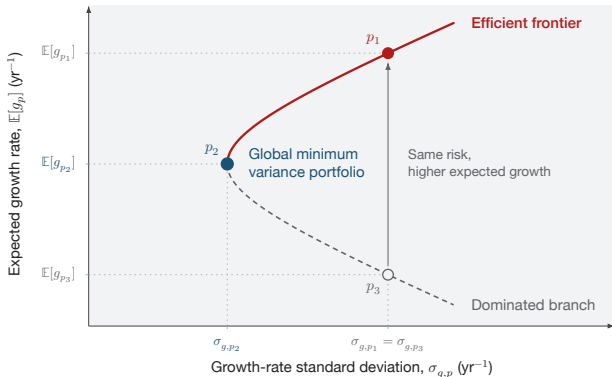
The minimum variance at the target is:

$$\sigma_g^2(g_\star) = \frac{ag_\star^2 - 2bg_\star + c}{d} = \frac{1}{a} + \frac{a}{d} \left(g_\star - \frac{b}{a} \right)^2.$$

Varying $g_\star$ traces the **minimum-variance frontier**. At $g_\star = b/a$, we recover the GMV variance $1/a$; higher or lower targets require more variance.

The Efficient Branch

A portfolio is efficient if no feasible alternative offers at least as much expected growth and no more variance, with one strict improvement.



- p_2 is the GMV portfolio.
- p_1 and p_3 have the same risk; p_1 has higher growth.
- The upper branch, including p_2 , is efficient.

Feasible portfolios lie on or to the right of the frontier; moving right adds variance.

Tracing the Long-Only Frontier

Long-only portfolio growth is a weighted average of the sample means:

$$\min_i g'_i \leq (\mathbf{g}')^\top \mathbf{w} \leq \max_i g'_i.$$

This gives a practical target sweep:

- Start with $g_\star \leq \min_i g'_i$. Every long-only portfolio meets the floor, so the solution is the long-only GMV portfolio.
- Increase the target to trace the efficient branch. Targets above $\max_i g'_i$ are infeasible.
- With the same inputs, long-only constraints cannot reduce variance. The frontiers meet where the shorts-allowed weights are nonnegative.

Let's estimate the inputs, sweep the target, and inspect how the weights change.

Example 1: [Minimum-Variance Portfolios from Data](#)

The Capital Allocation Line

A risk-free asset has known growth g_f (year⁻¹) and zero variance and covariance. A Treasury bill maturing at the holding period's end is one example.

Put fraction w_f in this asset and $1 - w_f$ in risky portfolio p , with $\sigma_{g,p} > 0$. The **complete portfolio** c has:

$$\mathbb{E}[g_c] = g_f + (1 - w_f)(\mathbb{E}[g_p] - g_f),$$

$$\sigma_{g,c} = |1 - w_f| \sigma_{g,p}.$$

For $w_f \leq 1$, substitute $1 - w_f = \sigma_{g,c}/\sigma_{g,p}$ to obtain the **capital allocation line (CAL)**:

$$\mathbb{E}[g_c] = g_f + \underbrace{\frac{\mathbb{E}[g_p] - g_f}{\sigma_{g,p}}}_{\text{Sharpe ratio } \text{SR}_p} \sigma_{g,c}.$$

Each risky portfolio defines a line through $(0, g_f)$ with slope SR_p .

Which Sharpe Ratio?

The CAL slope $SR_p = (\mathbb{E}[g_p] - g_f)/\sigma_{g,p}$ is dimensionless: both numerator and denominator have units year^{-1} . It is the Sharpe ratio for **one observation interval**.

Under independent increments with constant parameters, summing interval log returns multiplies the mean by the number of intervals and the standard deviation by its square root. For $\tau = 1$ year:

$$SR_p^{\text{annual}} = \sqrt{\frac{\tau}{\Delta t}} SR_p = \frac{\sqrt{\tau} (\mathbb{E}[g_p] - g_f)}{\sqrt{\mathbf{w}^\top \mathbf{C} \mathbf{w}}}.$$

Daily observations have $\Delta t = 1/252$ years, so the annualized ratio is $\sqrt{252} \approx 15.9$ times the one-day ratio.

Both conventions rank portfolios identically. L6b onward reports the annualized value.

The Tangent Portfolio

The **tangent portfolio** $\mathcal{T}$ maximizes the Sharpe ratio and gives the steepest CAL. For the closed form, allow shorts, assume Σ_g is positive definite, and require GMV expected growth to exceed g_f .

Define excess growth $\mathbf{e} = \mu_g - g_f \mathbf{1}$. Setting the Sharpe-ratio gradient to zero gives:

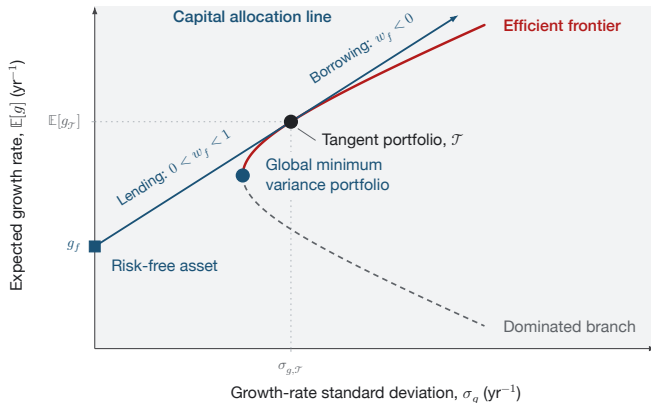
$$\Sigma_g \mathbf{w} = \frac{\mathbf{w}^\top \Sigma_g \mathbf{w}}{\mathbf{e}^\top \mathbf{w}} \mathbf{e}, \quad \mathbf{w} \propto \Sigma_g^{-1} \mathbf{e}.$$

Normalize the weights to sum to one:

$$\mathbf{w}_{\mathcal{T}} = \frac{\Sigma_g^{-1}(\mu_g - g_f \mathbf{1})}{\mathbf{1}^\top \Sigma_g^{-1}(\mu_g - g_f \mathbf{1})}.$$

With long-only weights, solve numerically. The data example approximates $\mathcal{T}$ by selecting the largest Sharpe ratio along its frontier sweep.

Lending and Borrowing on the CAL



- **Lending**, $0 < w_f < 1$: split wealth between the risk-free asset and $\mathcal{T}$.
- **Fully risky**, $w_f = 0$: hold $\mathcal{T}$ alone.
- **Borrowing**, $w_f < 0$: borrow at g_f to invest more in $\mathcal{T}$.

Lending and borrowing use the same rate g_f . Changing w_f moves along the CAL while keeping the same risky fund.

Two-Fund Separation

The two funds are the **risk-free asset** and the **tangent portfolio** $\mathcal{T}$. Investors make two separate choices:

1. **Choose the risky asset mix.** Mean-variance investors using the same inputs and investment opportunities choose the same $\mathcal{T}$, which maximizes the Sharpe ratio.
2. **Choose how much to invest in it.** Each investor's risk tolerance determines w_f , the fraction in the risk-free asset. The remaining fraction $1 - w_f$ goes into $\mathcal{T}$.

Risk preferences change the amount invested in $\mathcal{T}$; the asset proportions within $\mathcal{T}$ stay the same.

This assumes a common rate for lending and borrowing (Tobin, 1958).

Borrowing and Portfolio Choice

With \$100 of our own wealth, a \$150 investment in $\mathcal{T}$ requires a \$50 loan ($w_f = -0.5$).

What happens when the borrowing rules change?

- **No borrowing:** the \$150 position is unavailable. We can invest at most our \$100 in $\mathcal{T}$. To seek more growth than $\mathcal{T}$ offers, we must choose a different risky asset mix.
- **Borrowing costs more than lending earns:** the same \$150 position has higher financing costs, so it gives less expected net growth at the same risk. We must choose the risky mix using that borrowing cost.

Different borrowing opportunities can therefore lead investors to choose different risky funds.

Estimating Portfolio Inputs

The GMV calculation uses the estimated covariance $\hat{\Sigma}_g$. The tangent calculation also uses the estimated mean growth rates $\mathbf{g}'$. The weights therefore depend on the data and estimation method.

- For AMD (2014–2024), mean growth estimates are 0.3120 per year from the sample mean and 0.4397 per year from L4b's regression: a difference of 12.8 percentage points per year.
- Underestimating an asset combination's variance can make the optimizer favor it.

Our data example evaluates the selected allocations on 2025 prices, alongside equal weights and an index fund.

Example 2: [Portfolio Wealth with Multiple Asset GBM](#). Hold the fitted inputs and allocation fixed; simulate buy-and-hold wealth and estimate the probability of exceeding a target scaled NPV.

Optional Advanced Material

These examples develop the frontier geometry and input sensitivity further.

Advanced 1: Frontier Geometry and the Two-Fund Theorem

How can two risky frontier portfolios generate the rest of the curve? Derive the weights and vary the mix of two fixed funds. Compare the frontier with a 50% limit on each position's magnitude and with long-only weights.

Advanced 2: How Estimates Affect Portfolio Weights

How much do weights change when we estimate the inputs again? Resample the 2014–2024 observations, vary means and covariance separately, and compare GMV and tangent allocations. Evaluate the resulting long-only portfolios on the same 2025 prices.

Summary

We formulated portfolio reward and risk, constructed minimum-variance portfolios, and added a risk-free asset.

Key takeaways

- **Portfolio reward and risk:** expected growth is linear in the weights; variance includes covariance. For two assets with positive weights, correlation below one reduces risk below the weighted average of their standard deviations.
- **Minimum-variance portfolios:** GMV gives minimum variance without a growth constraint. Higher feasible growth targets trace the efficient branch; restrictions on the weights change the frontier.
- **The risk-free asset and tangent portfolio:** the CAL slope is a Sharpe ratio, maximized by the tangent portfolio. Two-fund separation lets investors vary risk through the risk-free allocation.

Next: model asset growth using a shared market factor and firm-specific risk.